

SEION Math Core v4 Supplementary Atlas

Eliuth Chavero Jasso
Independent Researcher, Apizaco, Tlaxcala, Mexico

29 July 2026

Abstract

This atlas records the finite-dimensional evidence, artifact contracts, negative controls, and reproducibility boundaries associated with the canonical SEION Math Core v4 research draft. It is a companion record, not an independent proof of universal or continuum statements.

1 Scope and authority

The mathematical paper separates definitions, conditional results, observed numerical residuals, and unresolved claims. The authority ladder is recorded in `governance/AUTHORITY_LADDER.yaml`; the machine graph and source hashes are in `.ai/machine`.

2 Canonical artifact map

Artifact family	Authority
Theorem/proof	theorem registry and proof source
Executed result	run manifest, metrics, certificate, hashes
Experiment design	registered experiment matrix
Figure/table	generated output with provenance
Release status	independent gate report and human decisions

3 Required controls

The registry distinguishes known invariant projectors, random projectors, PCA/SVD, spectral projectors, and closure-minimizing projectors. Exact dense laws and CP rank sweeps are controls; negative spectral-gap controls remain mandatory. Repeated executions are deduplicated by scientific-instance identity and are not counted as independent experiments.

4 Mathematical blockers

The v4 program does not silently promote the current tree-level certificates to a universal theorem. Sharpness extremizers, a complete prior-art novelty audit, cancellation-aware FI/GJI/Jacobiator constants, and a full extended optimizer grid remain explicit blockers until independently completed.

5 Reproducibility manifest

```
python -m pytest -q
python scripts/build_v4_foundation.py
python scripts/seion_campaign_v4.ps1
```

The last command is a PowerShell orchestrator; all stages report status and preserve failed evidence.

6 Selected figures

Figures are generated from registered v3 runs and copied into the v4 paper tree. Their captions state the metric, seed count, baseline, and limitation. No categorical dtype progression is treated as a continuous variable.

7 Reviewer checklist

- Claims map to theorem, experiment, counterexample, or blocker records.
- No numerical residual upgrades a claim to **PROVED**.
- Tables obey lower/upper and operator/Frobenius metric invariants.
- PDF and source checksums are emitted by the release script.
- Human review remains a required external decision.